A General Procedure to Combine Estimators

Statistics for Data Science
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Introduction

- Several estimators can be used for the same parameter
- No single estimator is always better: performance depends on sample size n and on the true model
- Combining them achieves better performances than any single one
- Combination through **weighted average** of estimators:

$$\hat{\theta} = \lambda^{\intercal}\mathbf{T}, \lambda \in \Lambda$$

Averaging Procedure

- The target is to approximate the model's weights to the ones used for averaging by the oracle estimator
- The oracle is the optimal linear combination of the available estimators, i.e. the configuration that achieves lowest mean squared error:

$$\lambda^* = \arg\min_{\lambda \in \Lambda} \mathbb{E}(\lambda^\intercal \mathbf{T} - \theta)^2$$

- Weights are optimized by minimizing the Mean Squared Error

Definition of the Constraints set

Approaches for defining the weight constraints set Λ:

- **Larger admissible set**
  - Better performances
  - Higher complexity
- **Smaller admissible set**
  - Potentially Less efficient
  - Simpler and more Stable

Common choice: weights sum up to 1 (used for replicating all experiments shown later)

Estimation of MSE matrix Σ

The oracle can be rewritten using the MSE matrix of the estimators, denoted by Σ:

$$\lambda^* = \arg\min_{\lambda \in \Lambda} \mathbb{E}(\lambda^\intercal \mathbf{T} - \theta)^2 = \arg\min_{\lambda \in \Lambda} \lambda^\intercal \Sigma \lambda$$

In the case where the weights sum to one, the minimization problem admits a *closed-form* solution depending only on Σ:

$$\lambda^*_{\max} = \frac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}^\intercal \Sigma^{-1}\mathbf{1}}$$

Problem: Σ is unknown and must be estimated from data; strategies depend on the model's assumptions

Application 1 - Symmetric Distribution

Estimates the location parameter θ of a symmetric distribution

Target distributions used:

- Gaussian
- Logistic
- Mixture
- Student-*t* (df = 4 and 7)
- Cauchy

Candidate estimators:

- Mean
- Median

Application 1 - Procedure

Estimates of the MSE matrix:

1. Plug-in method that produce the estimator AV
2. Bootstrap method that produce the estimator AVB

Sample size: 30, 50, 100

Number of replication: 10^4

Number of bootstrap sample: 1000

Application 1 - Results

	MSE on 30 observations				MSE on 100 observations			
Target distribution	**Mean**	**Median**	**AV**	**AVB**	**Mean**	**Median**	**AV**	**AVB**
Student(4)	0.10	0.08	0.08	0.08	0.03	0.02	0.02	0.02
Logistic	0.15	0.18	0.15	0.15	0.05	0.06	0.05	0.05
Student(7)	0.07	0.08	0.07	0.07	0.02	0.02	0.02	0.02
Gaussian	0.05	0.07	0.05	0.05	0.01	0.02	0.01	0.01
Mixture	0.24	0.83	0.24	0.31	0.07	0.43	0.05	0.06
Cauchy	6192328	0.14	522631	0.14	1036584.16	0.04	40127.16	0.03

Application 2 - Weibull

Model: X~ Weibull (β, η) with true parameters **β = 2, η = 10**

Sample size: n = 20

Estimators:

- Shape **β**: ML (Maximum Likelihood), MM (Method of Moments), OLS (Ordinary Least Squares)
- Scale **η**: ML

Averaging:

- Applied only to the estimators of **β**
- $\eta_{AV} \approx \eta_{ML}$

Application 2 - Results

Numerical Results (Monte Carlo, R = 300, B = 300):

- MSE (β):

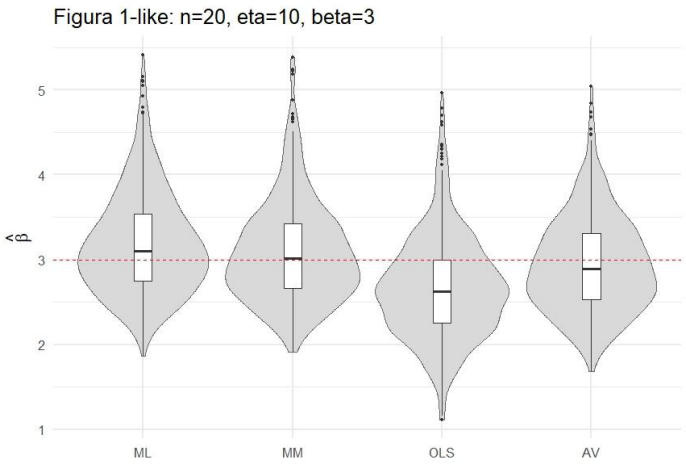
  ML = 0.179, MM = 0.147, OLS = 0.190, ***AV = 0.141***

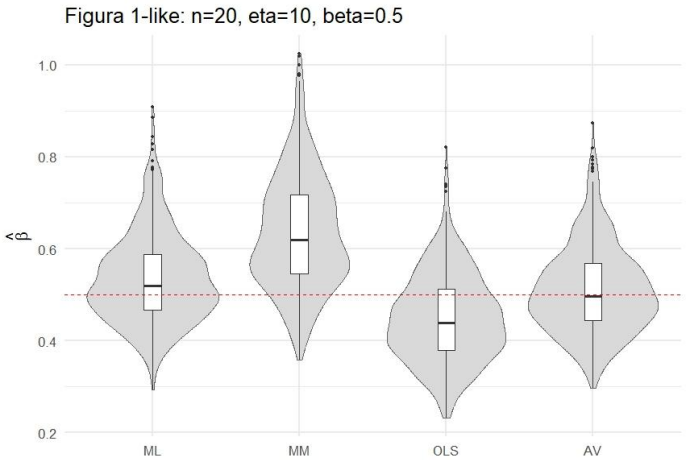
- MSE (η):

  ML = 1.469, ***AV = 1.455***

- Coverage: ≈ 93% for both β and η

AV estimators more centered on the true values





Application 3 - Estimation under misspecification

Estimates the *p-th* quantile of the target distribution

Estimators:

- Nonparametric estimator
- Weibull distribution-based estimator
- Gamma distribution-based estimator
- Burr distribution-based estimator

Application 3 - Original models

Quantile to estimate: ***p* = 0.99**

Target distributions used:

- Weibull distribution with parameters **α=3**, **β=2**
- Gamma distribution with parameters **α=3**, **λ=2**
- Burr distribution with parameters **β=2**, **η=1**
- Standard lognormal distribution (**All used estimators misspecified!**)

Number of observations: 100, 1000

Application 3 - Results

	100 observations			1000 observations		
Target distribution	**Correct Dist. MSE**	**NP MSE**	**AV MSE**	**Correct Dist. MSE**	**NP MSE**	**AV MSE**
Weibull (*1000)	23.25	55.17	48.13	2.13	5.65	2.69
Gamma	0.11	0.34	0.23	0.01	0.04	0.04
Burr	9.18	20.64	13.35	0.80	2.15	8.67
Lognormal	-	10.54	11.22	-	1.68	9.61

Results for 200 Monte-Carlo, 200 Bootstrap

Conclusions

The averaging procedure is not only a practical idea, but is supported by solid theoretical results:

- *Non-Asymptotic Bound:*
  - It remains close to the Oracle even in small samples, if the covariance matrix is well estimated.
- *Asymptotic Bound:*
  - For large samples, it is as efficient as the Oracle

Overall, averaging is stable and often strictly better

Thank you for your attention